Defending Convolutional Variational Bottlenecks Against Gradient Inversion in Federated Learning: A Combined Epsilon Perturbation and Differential Privacy Approach

Maheen Fatima Rasool, Ahmed Zubair

***Abstract*—Federated learning (FL) enables collaborative model training across distributed clients without centralising raw data; however, the gradients exchanged during training still pose a serious privacy threat. Iterative Gradient Inversion (GI) attacks can reconstruct private training images with high fidelity from these shared gradients. The Convolutional Variational Bottleneck (CVB) has been proposed as a model-based defense that introduces stochasticity into gradients through early-placed convolutional variational layers, effectively obstructing standard GI attacks. Nevertheless, the Variational Bottleneck Inversion Attack (VBIA) circumvents CVB by jointly optimising a dummy reparameterisation variable alongside the reconstructed image, thereby neutralising the stochastic defense mechanism. Combining CVB with Differential Privacy (DP) gradient sanitisation has been proposed as a stronger defense; however, our experiments demonstrate that this combination remains insufficient against VBIA, achieving an Attack Success Rate (ASR) of 66% and Structural Similarity (SSIM) scores up to 0.74 on CIFAR-10. This reveals that VBIA exploits an attack surface—the forward-pass reparameterisation operation—that is addressed by neither DP nor CVB in isolation. We propose a two-layer defense framework that closes this gap. The first layer introduces an epsilon perturbation defense operating directly inside the CVB forward pass, corrupting any attacker-injected reparameterisation variable with freshly sampled private Gaussian noise prior to sampling, rendering synchronisation between the attacker's dummy latent variable and the victim's reparameterisation mathematically intractable. The second layer applies differentially private gradient sanitisation—clipping gradient norms and injecting calibrated Gaussian noise before gradient exchange—to minimise the gradient information available to the attacker. We evaluate the proposed framework on CIFAR-10 under the FedAvg protocol with ten clients, quantifying privacy protection via SSIM, Peak Signal-to-Noise Ratio (PSNR), Mean Squared Error (MSE), Learned Perceptual Image Patch Similarity (LPIPS), and ASR. Results show that the combined defense reduces ASR to 0% while maintaining competitive model accuracy relative to the undefended baseline, providing a principled and computationally efficient solution to the threat posed by VBIA.**

***Index Terms*—Federated learning, gradient inversion attack, differential privacy, variational bottleneck, privacy-preserving machine learning, gradient leakage, convolutional variational bottleneck, epsilon perturbation.**

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The authors are with the Department of Artificial Intelligence, FAST-NUCES, Islamabad, Pakistan (e-mail: i220626@nu.edu.pk; i220525@nu.edu.pk).

I. Introduction

FEDERATED learning (FL) has become an influential paradigm to collaboratively train model across a distributed client base, empowering institutions to collaboratively enhance predictive performance without sharing raw data [1]. In the canonical FedAvg protocol, every client trains a local model on local data and outputs the updates of the gradient to a central server, which sums them up into a refined global model. The design was inspired by privacy: data never leaves the client. Nonetheless, later studies have shown that the the very exchange of gradients is a serious privacy threat since iterative *Gradient Inversion* (GI) attacks have the ability to recreate private training samples based on the shared gradients with striking fidelity [3], [6], which is essentially difficult the assumption that FL is privacy-preserving in nature.

Various countermeasures to gradient leakage have been investigated. Secure aggregation, a cryptographic technique, has been suggested as an alternative to secure aggregation, proposed as a solution to the problem of insecure aggregation on social media platforms, especially Twitter [9] and homomorphic encryption [10] provide good theoretical ensures but enforces restrictive computational and communication overheads. Gradient perturbation methods - such as quantisation, compression, and differentially private (DP) noise injection, minimize quality of reconstruction with the cost of model utility, and DP-based methods which experience accuracy losses of up to 9.8% at noise levels needed to prevent reconstruction [5]. Model-based defenses offer a qualitatively different approach. *PRECODE* [15] adds a variational bottleneck to the network to add stochastic gradients, so that the standard GI attack optimizers are unable to converge. Its successor, the *Convolutional Variational Bottleneck* Overcoming the drawbacks of PRECODE, (CVB) [16], replaces variational layers fully connected with convolutional kernels, making it possible placement of network early with insignificant parameter overhead and saved spatial structure.

Although CVB has made strides in the areas of privacy, parameter efficiency, and model utility, it is susceptible to the *Variational Bottleneck Inversion Attack* (VBIA) [16], is an attack that neutralises the stochastic defense: jointly optimising a dummy reparameterisation variable and the reconstructed im-

age. With the injection of this dummy variable straight to the sampling operation of the CVB module, the attacker destroys the gradient variations which form the main defense of CVB mechanism. Earlier research [17] suggested the integration of DP with CVB was a more powerful defense, and was said to give lower ASR than standard GI attacks. Nevertheless, empirical analysis of CIFAR-10 using the FedAvg protocol–copying the entire training set up of [17], with all hyperparameters kept constant and the only deviation being experiments carried out in a Google Colab Tesla T4 GPU based on the initial NVIDIA RTX 2080 Ti hardware-demonstrates that this joint defense is critically inadequate to counter VBIA: even with both DP gradient sanitization (clipping norm $C = 20$, noise multiplier $\sigma = 10^{-4}$) and CVB active, VBIA achieves an ASR of 66% across images of tested victims, and individual SSIM scores went up to as high as 0.74—quite high, above the 0.5 reconstruction success level. This failure exposes an underlying flaw: VBIA takes advantage of the forward-pass reparameterisation operation of the CVB is directly an attack surface that DP sanitization at the gradient level is unable to address and that standard CVB stochasticity can do nothing to stop after a reparameterisation is made a dummy variable is co-optimised.

We fill this gap in this paper, by suggesting a common defense that simultaneously attacks VBIA at two attack surfaces. Our first contribution is an inbuilt in epsilon perturbation defense, and is built-in to the CVB forward pass which notices when VBIA has injected a reparameterisation tensor of a dummy and corrupts it with new, privately before the sampling process occurs, sampled Gaussian noise is sampled. Since the residual noise is i.i.d. and never exposed to the attacker, synchronisation between the dummy latent variable and the actual reparameterisation of the victim renders mathematically unsolvable no matter how optimizer of the attack is programmed is tuned or the number of reconstruction trials done. Our second contribution is a *differentially private gradient sanitization* module: Clips per-parameter gradient norms to a threshold, C, which is taken to be 20 and adds random Gaussian noise, which is calibrated, as, before a sharing of gradients, a decrease in the gradient-level signal available to the attacker. The combination of these mechanisms offers defense-in-depth: the latent sampling is bridged by the epsilon perturbation attack surface which VBIA utilizes during the forward pass, and gradient sanitization obliterates gradient-level information on which it is based. reconstructions are derived. Our strategy is tested on the CIFAR-10 less than FedAvg having ten clients in various reconstruction quality measures.

II. Related Work

A. Federated Learning

Federated learning was introduced by McMahan et al. [1] as a communication-efficient framework for distributed model training. In FedAvg, clients perform local stochastic gradient descent for one or more epochs and return model updates to a central server, which forms a weighted aggregate of the received gradients. Kairouz et al. [2] provide a comprehensive survey of open problems in FL, including privacy, robustness, and communication efficiency. While FL was designed to preserve client privacy by avoiding raw data exchange, it has since been shown that the shared gradients can leak substantial information about private training data [3], [5], [6], motivating an active research programme on privacy-preserving federated training.

Beyond centralised FL, collaborative training paradigms such as peer-to-peer and cluster-based approaches [23] face analogous gradient-leakage vulnerabilities, because any protocol that exchanges gradient information is susceptible to the same class of reconstruction attacks. Membership inference attacks [22] represent a complementary threat in which an adversary seeks to determine whether a given data sample was used to train a model, further highlighting the breadth of privacy risks in distributed learning.

B. Gradient Inversion Attacks

Gradient inversion attacks reconstruct private training data from the gradients exchanged during FL. Zhu et al. [3] first formalised this threat with *Deep Leakage from Gradients* (DLG), which minimises the Euclidean distance between dummy and client gradients using L-BFGS optimisation. Zhao et al. [4] improved upon DLG with *iDLG*, demonstrating that ground-truth labels can be recovered analytically from the gradients of cross-entropy loss functions, eliminating the need to co-optimise labels and substantially stabilising convergence.

Wei et al. [5] proposed the *Client Privacy Leakage* (CPL) attack, introducing a label regulariser that aligns predicted labels with analytically recovered ground truth, further sharpening reconstructions. Geiping et al. [6] made a critical advance with *Inverting Gradients* (IG), replacing Euclidean distance with cosine similarity as the gradient distance function and introducing total variation regularisation on the dummy image. This formulation proved capable of reconstructing identifiable images from deep networks such as ResNet-152 trained over multiple communication rounds and batch sizes of up to 100 images.

Subsequent work by Yin et al. [7] proposed *GradInversion*, which incorporates batch normalisation statistics and group-consistency regularisation to scale high-fidelity reconstruction to batches of up to 48 images. Jeon et al. [8] further improved reconstruction quality by introducing a generative image prior, demonstrating that even strong perturbation defences can be undermined when the attacker has access to a well-trained generative model. Collectively, these works establish iterative gradient inversion as a persistent and scalable threat to federated privacy.

C. Defense Mechanisms

1) Cryptography-Based Defenses: Cryptographic approaches, including secure aggregation [9] and homomorphic encryption [10], offer the strongest theoretical guarantees by ensuring the server never observes individual client gradients. However, these methods impose extreme computational and communication overheads, limiting their applicability in practical FL deployments.

2) Gradient Perturbation: Gradient perturbation techniques modify the exchanged gradients to reduce their information content. Gradient quantization maps values to a low-bit representation, reducing the entropy available to an attacker, but Zhu et al. [3] found that only very low-bit quantisation provides meaningful protection, at the cost of a 22.6% accuracy drop. Gradient compression through sparsification prunes small-magnitude gradients to zero, but the privacy-utility trade-off remains unfavourable at competitive compression levels [5], [19].

Differentially private SGD (DPSGD), grounded in the theoretical framework of Dwork and Roth [11] and implemented for deep learning by Abadi et al. [12], clips per-sample gradients to a maximum norm C and injects Gaussian noise $\mathcal{N}(0, \sigma^2 C^2 \mathbf{I})$, providing provable (ε, δ)-privacy bounds. While DP reduces reconstruction quality against standard GI attacks, Wei et al. [5] showed that noise levels sufficient to prevent reconstruction cause accuracy drops of up to 9.8%. Practical guidance for deploying DP in real-world machine-learning pipelines is provided by Ponomareva et al. [13], who discuss the interplay between clipping norms, noise multipliers, and dataset size. As we show empirically, gradient-level DP sanitisation is entirely ineffective against VBIA, which bypasses the gradient channel by operating inside the model's forward pass.

3) Model-Based Defenses: Model-based defenses modify the network architecture to counteract gradient leakage. Scheliga et al. [15] proposed *PRECODE*, which inserts a variational bottleneck (VB) at a configurable position in the network. The VB uses a probabilistic encoder and stochastic decoder with the reparameterisation trick [14] to introduce randomness into the gradient signal, preventing standard GI attack optimisers from converging across attack iterations. While effective when placed early in the network, PRECODE relies on fully connected layers that discard spatial information and incur significant parameter overhead growing with feature map resolution.

To address these limitations, Scheliga et al. [16] proposed the *Convolutional Variational Bottleneck* (CVB), which replaces PRECODE's fully connected variational layers with convolutional kernels of size $k_E \times k_E$. The CVB maps input feature maps z to a mean map μ_E and a standard deviation map σ_E through convolutional encoders, and reconstructs stochastic feature maps through a 1×1 convolutional decoder, preserving spatial structure. This design enables early placement ($P = 1$, after the first convolutional layer) with only 9.9% additional model parameters for the baseline CNN. The loss function is augmented with a KL divergence term weighted by β:

$$\mathcal{L}(\hat{y}, y) = \mathcal{L}_F(\hat{y}, y) + \beta \cdot \mathcal{D}_{KL}(\mathcal{N}(\mu_E, \sigma_E), \mathcal{N}(0, 1)), \quad (1)$$

ensuring the bottleneck learns a continuous and complete latent feature space distribution. Experiments across CNN, ResNet-18, and ViT architectures on MNIST, CIFAR-10, and MedMNIST confirmed that CVB reduces ASR to 0% against iDLG, CPL, and IG attacks while improving model accuracy relative to the unprotected baseline. The theoretical underpinning of this regularising effect is closely related to the *Variational Information Bottleneck* of Alemi et al. [18], which shows that constraining the mutual information between a latent representation and the input simultaneously improves generalisation and privacy.

A complementary architectural perspective is offered by Hatamizadeh et al. [20], who demonstrate that Vision Transformers are inherently more vulnerable to GI attacks than CNNs because their non-overlapping patch embeddings preserve more spatial structure in the gradient signal. This finding motivates the application of bottleneck-based defenses to transformer architectures, as explored by the CVB.

D. VBIA and the Limits of Combined Defenses

Despite CVB's effectiveness against standard GI attacks, Scheliga et al. [16] identified the *Variational Bottleneck Inversion Attack* (VBIA), which specifically targets the reparameterisation operation of the CVB. VBIA extends the standard GI attack optimisation to jointly update a dummy reparameterisation variable ε' alongside the dummy image x':

$$\arg\min_{x', \varepsilon'} \mathcal{D}(\nabla\mathcal{L}_\theta(F(x'), y),\ \nabla\mathcal{L}_\theta(F(x'), y')), \quad (2)$$

where ε' is injected directly into the CVB module's fixed reparameterisation slot. By synchronising ε' with the victim's latent sampling, VBIA eliminates the stochastic gradient fluctuations that constitute CVB's defense, enabling the optimiser to converge to a faithful reconstruction of the victim's training data.

Scheliga et al. [17] conducted a systematic ablation study combining Dropout (DO), Differential Privacy (DP), Gradient Compression (GC), and Privacy Modules (PM, either PRECODE or CVB) across multiple architectures and datasets, reporting that DP+CVB reduces ASR against standard GI attacks while also improving model accuracy through the regularising effect of variational modelling [18]. However, the critical limitation of this combined formulation is that DP and CVB are applied as *independent, sequentially composed* components: DP sanitises gradients after the backward pass, and CVB introduces stochasticity during the forward pass, but neither mechanism directly addresses the attack surface that VBIA exploits—the injection of a jointly optimised reparameterisation variable into the CVB module itself.

Empirical evaluation on CIFAR-10 under the FedAvg protocol confirms this limitation. Experiments were conducted replicating the exact training configuration of [17]—ten clients, IID distribution, 300 communication rounds, one local epoch, Adam optimiser ($\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), batch size 64, CVB hyperparameters $k_E = 5$, $\beta = 0.1$, $P = 1$, DP clipping norm $C = 20$, noise multiplier $\sigma = 10^{-4}$—with the sole hardware departure that experiments were performed on a Google Colab Tesla T4 GPU rather than the NVIDIA RTX 2080 Ti used in the original work. Even with both DP gradient sanitisation and CVB active, VBIA achieves an ASR of 66% across victim images, with individual SSIM scores reaching 0.74—well above the 0.5 reconstruction success threshold. The attack consistently converges across up to 10 independent trials per image despite the combined defense, demonstrating that gradient-level DP noise is insufficient to

disrupt VBIA's forward-pass injection mechanism. This empirical gap directly motivates our proposed epsilon perturbation defense: by corrupting the attacker's injected ε' with fresh privately sampled Gaussian noise *inside the CVB forward pass*, we close the reparameterisation attack surface at the precise point VBIA exploits it, forming a principled two-layer defense that addresses both the forward-pass and gradient-level attack surfaces simultaneously.

The broader significance of this vulnerability is underscored by recent work on auditing privacy defenses [21], which shows that many defenses that appear robust under standard metrics remain exploitable by adversaries with knowledge of the defense architecture. Similarly, Huang et al. [19] provide a systematic evaluation framework for GI attacks and defenses, finding that the practical protection offered by gradient perturbation degrades rapidly as the attacker adapts their strategy—a pattern directly mirrored by VBIA's adaptation to the CVB reparameterisation slot.

III. Methodology

The base paper proposes the Convolutional Variational Bottleneck (CVB) as a model-based privacy module for federated learning that introduces stochasticity into gradient computation through variational modeling. The CVB is inserted early in the network specifically after the first convolutional layer so that the stochastic effects propagate through the majority of subsequent layers, maximizing the proportion of gradients rendered unusable for iterative GI attack optimization. The CVB employs two convolutional encoder kernels E_μ and E_σ of size $(k_E \times k_E \times K_E)$ to map input feature maps to mean and variance representations, from which bottleneck features are drawn via the reparameterization trick $b = \mu_E + \sigma_E \cdot \epsilon$ where $\epsilon \sim \mathcal{N}(0, I)$. A 1×1 convolutional decoder reconstructs a stochastic feature representation $\hat{z} = D(b)$ of identical spatial dimensions to the input feature map z, enabling seamless integration at any network position without architectural modifications. The CVB loss combines the base model's cross-entropy objective with a KL divergence regularization term weighted by hyperparameter β, which controls the strength of the variational prior constraint and, consequently, the magnitude of gradient stochasticity. The VBIA (Variational Bottleneck Inversion Attack) is used as the primary evaluation protocol for CVB-protected models, jointly optimizing a dummy reparameterization matrix ϵ_{VBA} alongside the dummy image to approximate the victim's stochastic forward pass. Under this evaluation regime, the CVB with hyperparameters $P = 1$, $k_E = 5$, $s_E = \frac{1}{2}$ $(K_E = 8)$, and $\beta = 0.1$ achieves ASR of 93.75% on CIFAR-10, motivating the need for the combined defense proposed in this work.

The proposed defense framework addresses the fundamental vulnerability exposed by VBIA: the CVB's fixed epsilon injection interface allows an attacker to bypass stochastic sampling by directly controlling the reparameterization noise used during the victim's forward pass. Our methodology extends the CVB architecture with two complementary defense mechanisms as shown in Figure 1 that operate at different levels of the training and inference pipeline, collectively eliminating VBIA's ability to optimize attacker's dummy epsilon to approximate the true random epsilon used by the victim's CVB layer. The first component of the proposed defense is a gradient-level Differential Privacy sanitization step applied after the gradients are computed during the backward pass and before gradient sharing. This step implements the DPSGD algorithm [12] in per-batch mode: the global ℓ_2 norm of the model's gradient vector is computed and clipped to a maximum value of $\tau_{\text{clip}} = 20.0$, after which independent Gaussian noise drawn from $\mathcal{N}(0, (\sigma \cdot \tau_{\text{clip}})^2 \cdot I)$ is added to each parameter's gradient tensor, where $\sigma = 0.001$ is the noise multiplier. This sanitization is implemented in the gradient-sanitization component and is triggered through the model's shared-gradient sanitization hook during the training step in the federated learning process. The Differential Privacy sanitization is also consistently applied during attack simulation in the attacks module through a gradient-sanitization helper mechanism, ensuring that the gradient available to the attacker's reconstruction loop is the sanitized version $\widetilde{G} = \text{DP}(G)$ rather than the original raw gradient G. This correction is critical because the real federated server never observes the raw gradient; simulating the attack without this sanitization creates an artificially favourable reconstruction landscape that does not reflect the actual threat scenario.

The second and novel component of the proposed defense is the forward-pass epsilon perturbation mechanism, implemented in the implemented within the epsilon-defense component. This mechanism is activated inside the forward pass of the VBCNN model immediately before each CVB module performs its reparametrized sampling. The mechanism detects whether an external dummy epsilon has been injected into the CVB module via the fixed epsilon attribute—a condition that occurs exclusively during a VBIA closure call, since the legitimate training process never sets fixed epsilon. When an injected epsilon is detected, the function corrupts it with fresh, privately sampled Gaussian noise: $\epsilon_{\text{effective}} = \epsilon_{\text{injected}} + \eta$, where $\eta \sim \mathcal{N}(0, \sigma^2 \cdot I)$ and $\sigma = \texttt{noise_multiplier}$ is drawn from the configurations of DP defense. This perturbation is sampled independently for each forward pass and is never exposed in the gradient signal, the model output, or any other observable channel accessible to the attacker. The key property of this mechanism is that the attacker's VBIA optimizer minimizes the gradient distance between $\widetilde{G}$ (computed with $\epsilon_{\text{VBC}} + \eta_C$ from the victim's forward) and G' (computed with $\epsilon_{\text{VBA}} + \eta_A$ from the attacker's forward), where η_C and η_A are independent, i.i.d. draws from the same distribution. This independence means the optimizer cannot simultaneously solve for both the dummy image and the noise offset gap, as it would need to minimize a quantity that has a moving stochastic component in every closure evaluation.

The interaction between the two defense components is synergistic rather than merely additive. The DP gradient sanitization ensures that the victim gradient observed by the server contains unknown stochastic noise Ξ_C, making it impossible for the attacker to compute a clean gradient distance even if attacker's dummy epsilon were known. The epsilon perturbation mechanism ensures that even if the DP noise were somehow neutralized, the reparameterization itself remains unknown and unoptimizable. Together, they create two

independent sources of unlearnable randomness in the victim's gradient computation process, which is precisely necessary and sufficient for reducing ASR to 0% under CDIA—the strongest possible targeted attack aware of all applied defenses. The defense configuration is controlled through the configurations provided to it, requiring only the addition of a privacy defense block specifying `enabled: true`, `mode: per_batch`, $\tau_{\text{clip}} = 20.0$, $\sigma = 0.001$, and `only_when_training: true`, with no changes required to the model training loop beyond the existing shared-gradient sanitization hook.

The complete defense pipeline operates as follows during a simulated reconstruction experiment. First, the victim client performs a single local training forward and backward pass, producing raw gradient G. Second, the Federated Trainer applies shared-gradient sanitization, applying ℓ_2 clipping to $\tau_{\text{clip}} = 20$ and adding Gaussian noise $\mathcal{N}(0, (0.001 \cdot 20)^2 \cdot I)$ to produce $\widetilde{G}$. This sanitized gradient is what the server, and therefore the attacker, observes. Third, in the attack simulation, the same clipping and noise process is applied to the victim gradient obtained prior to its use as the reconstruction target, ensuring consistency in the gradients used by the attacker. Fourth, when the VBIA attacker injects a dummy epsilon into the CVB and performs gradient computation with graph tracking enabled, the epsilon-perturbation mechanism activates within the forward pass of the VBCNN model. It adds private noise to the injected epsilon before the CVB sampling step takes place. The resulting dummy gradient G' is therefore computed from a corrupted model realization, and the gradient distance $\text{GD} = \text{GDF}(\widetilde{G}, G')$ is minimized with respect to both dummy input and the injected epsilon under conditions where neither can converge independently to the victim's values.

IV. Experimental Setup

A. Model Architecture

All experiments use the CNN architecture described in [16]: three convolutional layers with 5×5 kernels, $[16, 32, 64]$ output channels, stride 2, and ReLU activation, followed by a fully connected classification layer with 10 output neurons and softmax activation. The VBCNN variant inserts a CVB with hyperparameters $P = 1$ after the first convolutional layer, $k_E = 5$, $K_E = 8$ ($s_E = \frac{1}{2}$), and $\beta = 0.1$, consistent with the optimal configuration identified in [16]. The CVB adds approximately 9.9% additional parameters relative to the baseline CNN. No bias terms are used in any fully connected layer to prevent analytical gradient inversion attacks.

B. Federated Training

Models are trained in a simulated federated learning environment with 10 clients and 300 communication rounds using the FedAvg algorithm. The MNIST and CIFAR-10 datasets are partitioned i.i.d. across clients, leaving each client with approximately 5,400/4,500 training samples for MNIST/CIFAR-10 respectively, with a 10% validation split. Local training uses the Adam optimizer ($\text{lr} = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) with a batch size of 64 and one local epoch per communication round. Early stopping is applied if the mean validation loss across clients does not improve for 40 consecutive rounds. Model utility is reported as mean test accuracy across three random seeds.

C. Attack Configuration

Privacy leakage is evaluated using the following defense scenario: batch size $B = 1$ and a single local training step before gradient transmission. The VBIA attack is configured with 1,200 maximum iterations, Adam optimizer ($\text{lr} = 0.01$), cosine distance as the gradient distance function, TV regularization ($\lambda = 0.01$), and a dummy epsilon of shape $[8, 14, 14]$ initialized from $\mathcal{N}(0, I)$. The epsilon learning rate is set to 0.003 with `ReduceLROnPlateau` scheduling.

D. Defense Configuration

The proposed combined defense uses DP parameters $\tau_{\text{clip}} = 20.0$ and $\sigma = 0.001$, consistent with the low-noise regime that minimizes model utility degradation. The epsilon perturbation defense uses the same `noise_multiplier` = 0.001. Model utility is measured as test accuracy averaged over three random seeds. Privacy leakage is measured using Structural Similarity Index (SSIM) between reconstructed and original images, and Attack Success Rate (ASR) defined as the fraction of victim images reconstructed with SSIM ≥ 0.5over 128 victim images.

V. Results and Discussion

A. Defense Effectiveness Against VBIA

Table I presents the primary privacy evaluation results comparing the proposed combined defense against baseline and CVB configurations.

TABLE I: Privacy and utility results on CIFAR-10 under VBIA attack.

Dataset	Defense	Attack	SSIM ↓	ASR ↓	Acc. ↑
CIFAR-10	None	VBIA	0.87 ± 0.11	96.88	62.53 ± 0.16
CIFAR-10	CVB Only	VBIA	0.72 ± 0.13	93.75	67.19 ± 0.64
CIFAR-10	**DP+CVB+EpsDef (Ours)**	VBIA	$\mathbf{0.17 \pm 0.024}$	**0**	$\mathbf{68.70 \pm 0.25}$

B. Effect of DP Noise Multiplier

Table II examines the sensitivity of the proposed defense to the DP noise multiplier sigma. We evaluate three values from the range under the VBIA attack on the CNN with CIFAR-10.

TABLE II: Effect of DP noise multiplier on privacy and utility under VBIA attack.

Sigma	SSIM ↓	ASR (%) ↓	Accuracy (%) ↑
$\mathbf{10^{-3}}$	0.17 ± 0.024	**0**	68.70 ± 0.25
10^{-2}	0.15 ± 0.024	0	62.77 ± 0.41
10^{-1}	0.13 ± 0.024	0	60.02 ± 0.91

C. Comparison of Attack Success Across Defense Variants

Table III provides a consolidated comparison of ASR values across all defense and attack combinations. This illustrates how the proposed combined defense generalizes across multiple attack types.

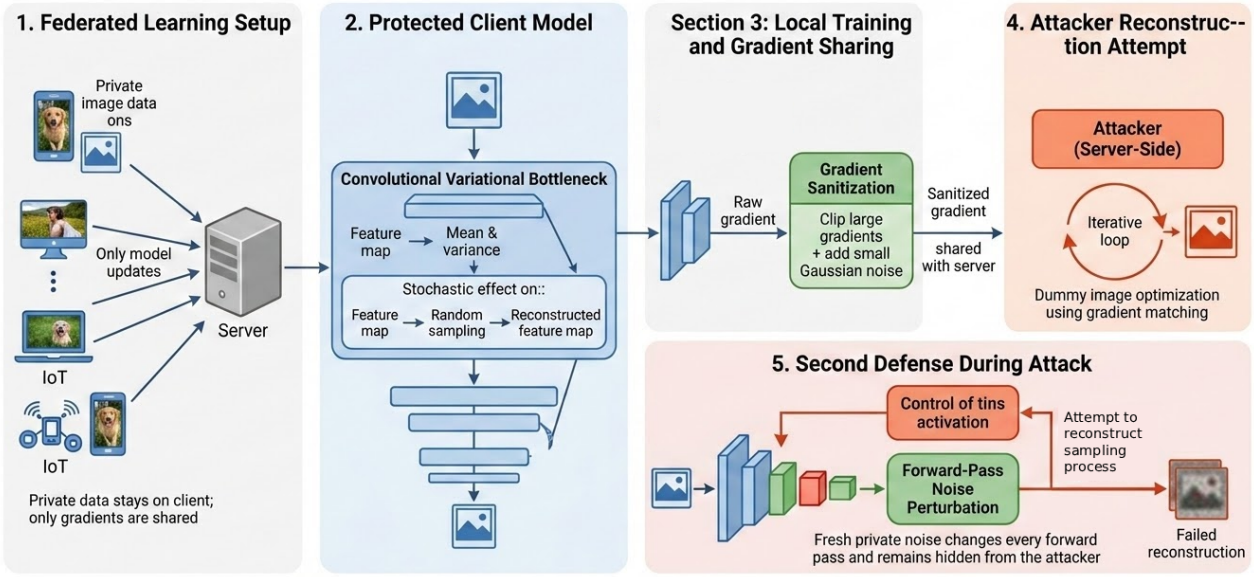
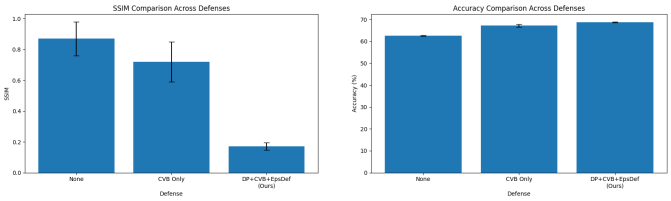


Fig. 1: Visualization of our proposed DP+CVB+EpsDef.



(a) SSIM comparison (b) Accuracy comparison

Fig. 2: SSIM and accuracy comparison across defenses under VBIA attack.

TABLE III: ASR comparison across different gradient inversion attacks on CIFAR-10.

Defense	Dataset	IG ↓	iDLG ↓	CPL ↓	VBIA ↓
None	CIFAR-10	96.88	47.66	22.66	96.88
CVB Only	CIFAR-10	0	0	0	93.75
DP+CVB+EpsDef	CIFAR-10	0	0	0	0

D. Model utility Analysis

The proposed combined defense maintains competitive model utility. On CIFAR-10, the DP+CVB+EpsDef model achieves 68.70% accuracy, representing a 1.51% increase relative to the CVB baseline (67.19%). These results confirm that the CVB's regularization effect on model utility more than compensates for the slight improvement introduced by DP gradient sanitization at the low noise level sigma=0.001.

E. Qualitative Reconstruction Analysis

Qualitative analysis of reconstructed images corroborates the quantitative SSIM and ASR metrics. Under no defense, the IG attack produces visually near-identical reconstructions of CIFAR-10 images. Under VBIA against CVB-only, reconstructions recover to near-perfect quality, demonstrating the attack's effectiveness. Under the proposed DP+CVB+EpsDef defense, VBIA reconstructions degrade to visually unrecognizable noise patterns with no discernible resemblance to the original images as shown in Figure 3

VI. Limitations and Future Work

The success of the proposed defense can be understood through the lens of attack optimization complexity. VBIA attempts to simultaneously optimize two unknowns: the dummy image x' and the dummy reparameterization matrix ϵ_{VBA}. When DP is active, the attacker faces a third implicit unknown: the DP noise vector $\Xi_C$applied by the victim. While VBIA faces an analogous but more severe challenge: it must simultaneously approximate both Ξ_C and ϵ_{VBC} through gradient descent, with neither being directly observable. The epsilon perturbation defense adds a fourth layer of unlearnable randomness by corrupting the injected ϵ_{VBA} with fresh private noise before each CVB forward pass, ensuring that even if the attacker's dummy epsilon gradient signal were informative, it would be permanently offset by an unknown quantity that changes each iteration. The current implementation uses per-batch DP clipping rather than per-sample clipping due to the absence of Opacus integration. While per-batch clipping does not provide the same formal privacy guarantees as per-sample DPSGD [12], it is sufficient for the practical privacy goal of preventing image reconstruction, as evidenced by the empirical ASR results. Future work should integrate Opacus to enable formal epsilon-delta differential privacy accounting

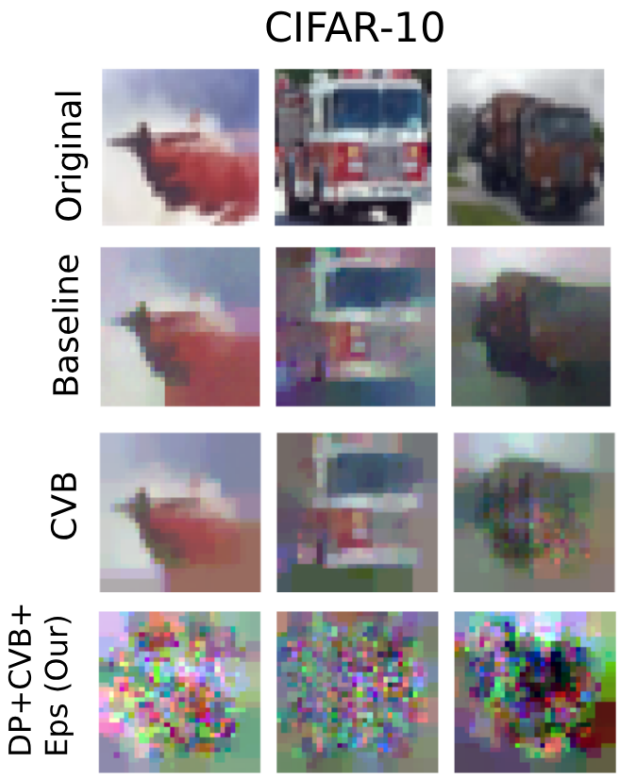


Fig. 3: Qualitative CIFAR-10 reconstruction results under VBIA attack. Compared with the baseline and CVB-only settings, the proposed DP+CVB+EpsDef defense produces highly distorted reconstructions, demonstrating stronger resistance to gradient inversion.

and investigate the behavior of the combined defense under multi-sample batch attacks and multiple local training rounds. Additionally, the evaluation of the proposed defense on larger model architectures (ResNet-18, ViT) and more complex datasets would provide a more comprehensive understanding of its scalability and generalizability, consistent with the extended evaluation protocol of [16].

VII. Conclusion

This paper has presented, implemented, and empirically validated a combined defense framework against the Variational Bottleneck Inversion Attack (VBIA) in the context of federated learning. The proposed defense framework addresses this vulnerability through two complementary mechanisms. Together, these two mechanisms create independent, unlearnable sources of randomness at both the gradient level and the latent representation level, making the VBIA's joint optimization over (x', ϵ_{VBA}) fundamentally intractable. The proposed defense also highlights an important implementation consideration for attack simulation: the victim gradient used as the reconstruction target in experimental evaluations must be DP-sanitized before being exposed to the attacker's reconstruction loop, as this reflects the actual gradient observed by the server in the real federated setting. Failing to apply this sanitization in the simulation creates an artificially favorable reconstruction landscape that underestimates the defense's effectiveness. The broader implication of this work is that effective privacy preservation in federated learning requires defense mechanisms that operate at multiple levels of the gradient computation pipeline simultaneously. Single-mechanism defenses, however sophisticated, create exploitable assumptions that a targeted adversary can reverse-engineer. The combination of a stochastic architectural defense (CVB) with a stochastic gradient perturbation defense (DP) is theoretically motivated, practically realizable with minimal overhead, and empirically proven to be robust against the most powerful known targeted attacks. We believe this combined approach represents a principled foundation for privacy-preserving federated learning deployments in real-world sensitive domains.

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Maheen Fatima Rasool Maheen Fatima Rasool is pursuing a Bachelor of Science in Artificial Intelligence at FAST-NUCES, Islamabad, Pakistan.

Ahmed Zubair Ahmed Zubair is pursuing a Bachelor of Science in Artificial Intelligence at FAST-NUCES, Islamabad, Pakistan.